

Ka-band high-rate telemetry system upgrade for the NASA deep space network[☆]

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ABSTRACT

The NASA Deep Space Network (DSN) has a new requirement to support high-data-rate Category A (Cat A) missions (within 2 million kilometers of the Earth) with simultaneous S-band uplink, S-band downlink and Ka-band downlink. The S-band links are required for traditional telemetry, tracking & command (TT&C) support to the spacecraft, while the Ka-band link is intended for high-data-rate science returns. The new Ka-band system combines the use of proven DSN cryogenic designs, for low system temperature, and high-data-rate capability using commercial telemetry receivers. The initial Cat A support is required for the James Webb Space Telescope (JWST) in 2014 and possibly other missions. The upgrade has been implemented into 3 different 34-meter Beam Waveguide (BWG) antennas in the DSN, one at each of the complexes in Canberra (Australia), Goldstone (California) and Madrid (Spain). System test data are presented to show that the requirements were met and the DSN is ready for Cat A Ka-band operational support.

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1. Introduction

Requirements for higher data rates from the NASA Earth and space science missions have created a need for wider channel bandwidths. This requirement has led to an evolution to Ka-band for the near-Earth (Category A) missions (within 2 million kilometers of the Earth). The NASA Space Network (SN) currently supports several Ka-band missions. The Near-Earth Network (NEN) currently supports the Lunar Reconnaissance Orbiter (LRO) and the Solar Dynamics Observatory (SDO) with a Ka-band 18 m antenna at White Sands (WS1), and has a long-range plan to add several 12 m S/Ka-band antennas [1]. While the NEN upgrade is primarily intended to support > 1 Gbps data rates from the low-Earth orbit (LEO), the Deep Space

Network (DSN) has a new requirement to support relatively high data rates (< 150 Mbps) out to the Sun–Earth Lagrange 1 and 2 (L1 and L2) points.

Several technology developments have been done at JPL and in industry in order to meet these new requirements. These new developments are mainly in the areas of the S/Ka dichroic plate, the broad-band LNA and the high-rate digital receiver. These technologies are described in the relevant paragraphs under section III.

Several 34-meter beam waveguide (BWG) DSN antennas have previously been upgraded to support the Category B (deep space) Ka-band, at 31.8–32.3 GHz, referred to as Ka1. However, this upgrade task, referred to as Ka-band Phase 2 or Ka2, is the first time the DSN will be used to support the Category A downlink band at 25.5–27.0 GHz. The S-band links are required for traditional Telemetry, Tracking and Command (TT&C) support to the spacecraft, while the Ka-band link is intended for high-data-rate science returns.

The advantages of the BWG antenna configuration for the DSN have been well documented [2]. The BWG

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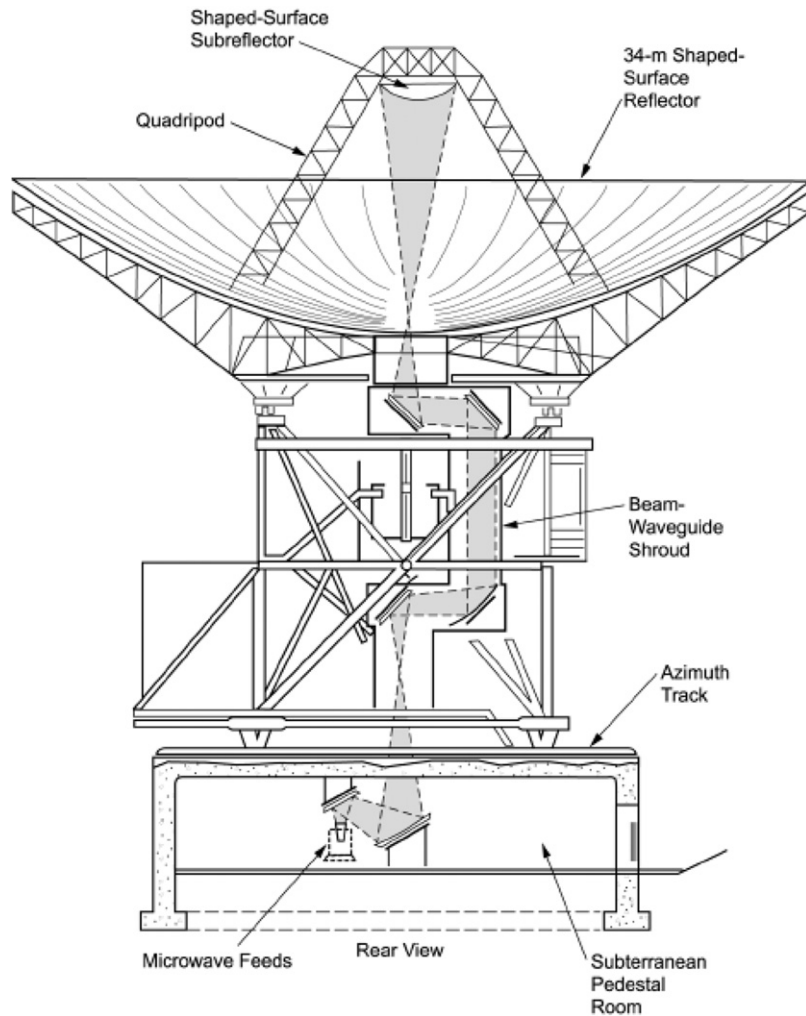


Fig. 1. 34 m Beam waveguide antenna with mirror configuration.

antenna construction (see Fig. 1) is easily adapted to multiple frequency bands, due to the space available in the pedestal room for the installation of additional mirrors and feeds. The addition of a new S/Ka dichroic mirror has enabled the simultaneous operation in the S and Ka2 bands. A photo of one of the upgraded antennas, the DSS-24 antenna at Goldstone, is shown in Fig. 2. The initial Cat A support is required for the James Webb Space Telescope (JWST) in 2014. TT & C Support for NASA lunar missions is also a possibility.

2. Requirements for Ka-band high-rate telemetry

The current mission requirements for the Ka2 upgrade dictate a low-noise, cryogenic system on the front end and a high-data-rate system on the back end. The system noise temperature is specified for a vacuum condition, i.e. not including atmosphere and cosmic background noise. In reality, the actual system temperature will be a strong function of weather, atmospheric conditions and antenna

elevation angle for the 26 GHz Ka2 frequency band [3]. The system is therefore designed to maintain the required 3 dB link margin for an atmospheric attenuation of up to 3 dB and a “weather” temperature increase of up to 100 K.

The key requirements for the Ka2 system are as follows:

Receive frequency band: 25.5–27.0 GHz.

Channel select step size: ≤ 1 MHz.

Receive input power range:

High-gain mode: -75 dBm, max.

Low-gain mode: -50 dBm, max.

RF system gain: (LNA input to telemetry receiver input).

High-gain mode: 57 ± 2 dB.

Low-gain mode: 37 ± 2 dB.

System non-linearity: $< 0.5\%$ (for -90 dBm ambient load noise input).

Link margin > 3 dB, for BER = 10^{-7} .

G/T ≥ 60.7 dB/K (vacuum) at 45 deg elevation.



Fig. 2. DSS-24 34 m antenna at Goldstone.

System noise temperature (vacuum) ≤ 36 K.
 Passband flatness: < 2.0 dB over 400 MHz.
 Receive polarization: RCP or LCP (not simultaneous).
 Antenna pointing accuracy < 5.5 mdeg (mean radial error (MRE)).
 Antenna gain measurement accuracy < 0.4 dB.
 Compression point dynamic range > 135 dB-Hz (in a 1 Hz bandwidth).
 Data rate: 1 to 150 Mbps, QPSK.
 Symbol rate: up to 350 Msym/s, QPSK.
 Modulation types: BPSK, QPSK, OQPSK.
 Waveform types: NRZ-L, M, S.
 Decoding formats: Viterbi (7, $\frac{1}{2}$), Reed Solomon.
 Local data storage: 5 Tbytes.
 Doppler measurement accuracy < 0.05 mm/s, 1 sigma.
 System noise temperature measurement accuracy: 5%.
 Data delivery probability: 95% (within 10 s of the Earth receive time (ERT)).

3. Technology development and system design

3.1. Optical design

The Ka2 task has added a new dichroic reflector (M6A) and flat plate mirror (M8), as well as a feed/LNA subsystem to enable simultaneous Category A S-band transmit, S-band receive and Ka-band receive. The notional mirror configuration is shown in the system block diagram in

Fig. 3. The new M6A mirror is selectable, using a shuttle mechanism, so that the antenna can operate in any of 3 modes (up/down/down): S/S/X, X/X/Ka1 or S/S/Ka2. For Ka2 band operation, the M7 mirror is moved out of the beam path into stow position.

The M6A mirror is a new broad-band design that is optimized for performance (< 10 K noise contribution across the band) and ability to be manufactured for relatively lower cost. Whereas previous large dichroic plates for the DSN have been fabricated using electrostatic discharge machining (EDM) to cut out the approximately 17,000 holes, which is time consuming and expensive, the M6A plate was designed such that it was feasible to be fabricated using a standard milling machine. This was made possible by increasing the inside radii of the square cutouts to 0.047 inches (from < 0.010 inches in previous designs) and relaxing the manufacturing tolerance from 0.0005 to 0.001 inches. A custom computer aided design (CAD) program was used to synthesize and then simulate the RF performance of the new design. The return loss test setup for the plate is shown in Fig. 4. The resulting return loss performance is shown in Fig. 5, showing very good agreement with the theoretical model. The null in the return loss response is placed near 25.9 GHz by design, since this is the downlink frequency of the primary mission (JWST).

3.2. Mechanical design

In order to add a feed under the new M8 mirror, it was necessary to build a new (Ka2) platform above the existing X/Ka platform. Since it is high enough not to interfere with any equipment movement in the pedestal room, and in order to be useful as a staging area, the platform was extended from the shroud to the basement wall (see Fig. 6). A modification was also made to the shroud by replacing a flat panel with a “dog house” panel, to allow the new M6A mirror to be withdrawn to the “stow” position. The M6/M6A shuttle function was implemented by extending the trolley rails for the original M6 and suspending the new M6A mirror on a new frame sitting at the other end of the rail system. The M6A mirror movement does not require any new motors since it is moved by being latched and towed by the original M6 mirror. The existing Feed Mirror Drive (FMD) software was modified to enable the latching and towing required for the new S/Ka2 mirror Configuration.

3.3. Cryogenic low-noise amplifier (LNA) subsystem

The low system temperature requirement of 36 K (in vacuum) was derived from the link budget required to support the relatively high data rates (up to 28 Mbps) for science missions at the edge of deep space, such as JWST at the L2 point. In order to achieve the low system temperature, it was necessary to employ a cryogenic system, cooled to about 10 K. The resulting LNA/feed effective temperature (T_e) is therefore about 14 K. The waveguide configuration inside the vacuum dewar (shown upside down), with the cryogenic coldhead attached, is shown in Fig. 7. This assembly includes the

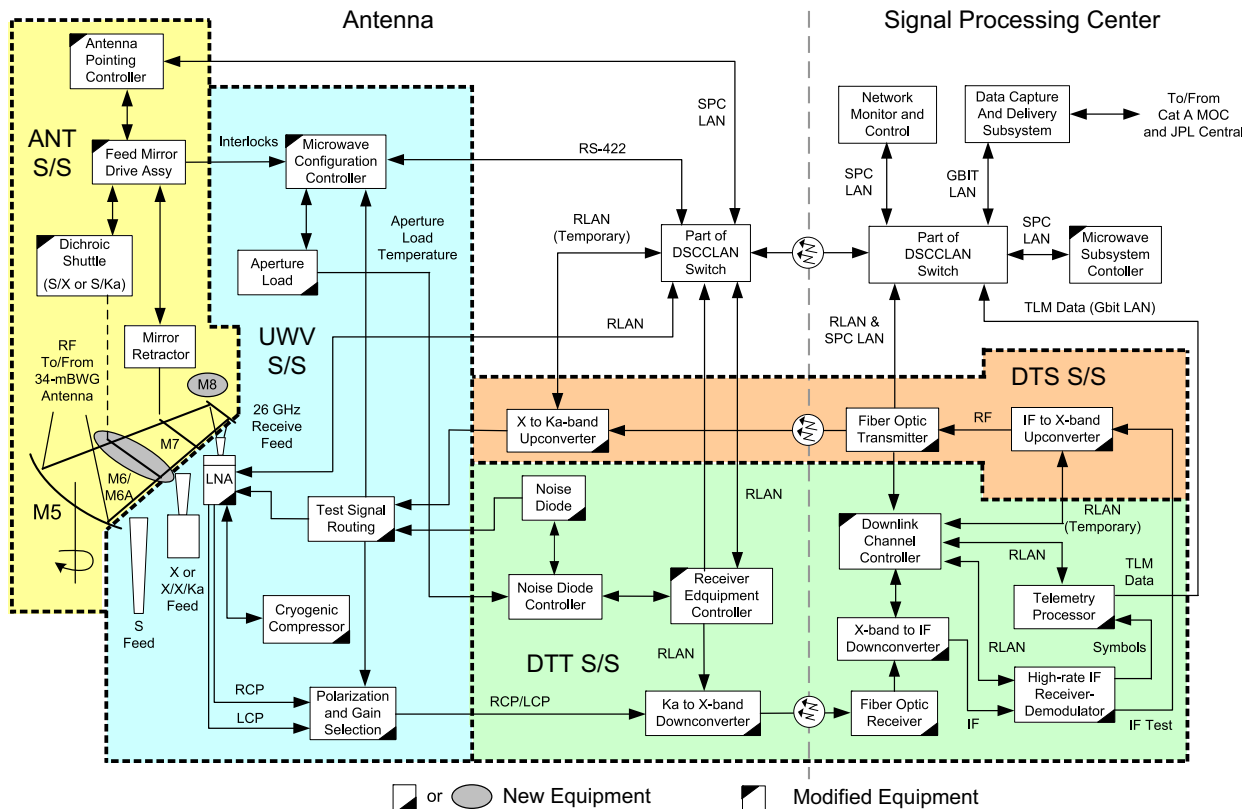


Fig. 3. Ka-band phase 2 system block diagram.

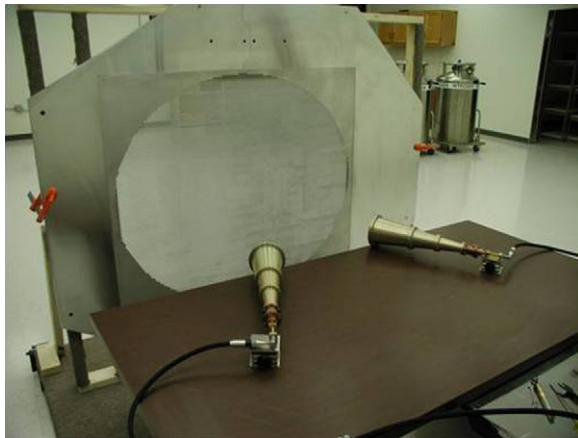


Fig. 4. S/Ka dichroic plate (M6A) test setup.

septum polarizer, couplers for noise and test signal injection and the LNA's for both RCP and LCP polarizations.

An RF plate assembly, mounted externally to the cryostat, is used for a post amplifier, polarization selection, test signal configuration and high/low gain selection. The low gain mode is 20 dB lower than the nominal 57 dB RF system gain and is used to support very strong signals during early post-launch mission phase. The existing

front-end controller software was modified to incorporate control of the new Ka2 equipment. A LAN interface was also added to remotely monitor the cryostat temperatures and pressures in the Signal Processing Centers (SPC).

The LNA module itself is a 4-stage amplifier, based on an Indium Phosphide (InP) high electron mobility transistor (HEMT) device and is shown in Fig. 8. The device itself is a legacy DSN design that has been used for the Ka1 32 GHz band, but the LNA matching circuit has been fine tuned to meet the ~ 11 K noise temperature requirement over the broader Ka2 band.

3.4. RF subsystem

The RF design is driven by both the high dynamic range and broad bandwidth requirements. The dynamic range requirement is achieved with a low-noise fiber optic link, to transport the X-band intermediate frequency (IF) signals from the antennas to the SPCs, which are up to 20 km apart. In order to minimize equipment in the antennas, a block Ka-to-X band downconverter is used in the antennas. Bringing the full 1.5 GHz bandwidth to the Signal Processing Center (SPC) also allows easy addition of more receiver channels, in the event that a requirement is added to support multiple spacecraft per antenna. This scenario is possible if there are multiple lunar missions operating simultaneously, for example. Further downconversion and channel selection, with an X-to-IF

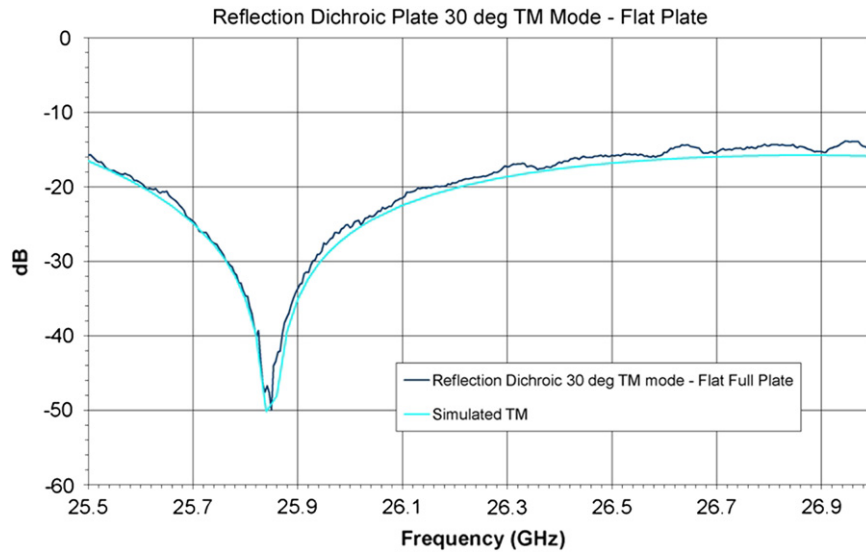


Fig. 5. M6A return loss—measured vs. theory.

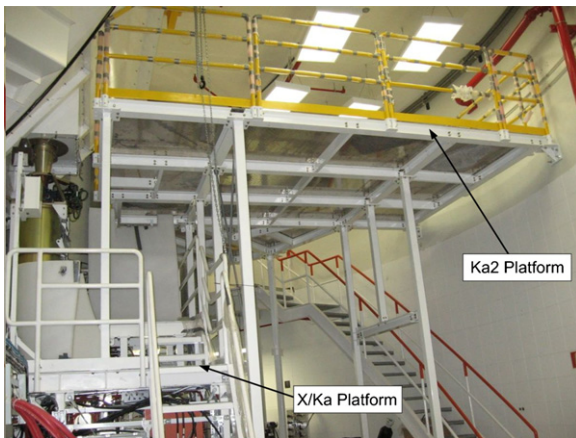


Fig. 6. Ka₂ platform in pedestal room.

downconverter, is then done in the SPC. An industry standard IF frequency of 720 MHz is used for compatibility with commercial telemetry receivers. Dual downconversion architecture is used in the synthesized downconverter to maintain a high spurious-free dynamic range in the presence of possible radio frequency interference (RFI).

System noise temperature (SNT) measurement is done remotely (via a LAN interface), using a synchronous detector circuit within the Ka–X block downconverter. The measurement approach is basically that of a noise-adding radiometer, using a modulated noise source and a synchronous Y-factor measurement. An ambient aperture load, mounted over the Ka₂ feed, is used to periodically calibrate the noise source.

Test signal generation is done using the internal test source of the RT Logic digital receiver. The 720 MHz IF test output is upconverted and tuned to the desired channel using the inverse architecture of the downconversion

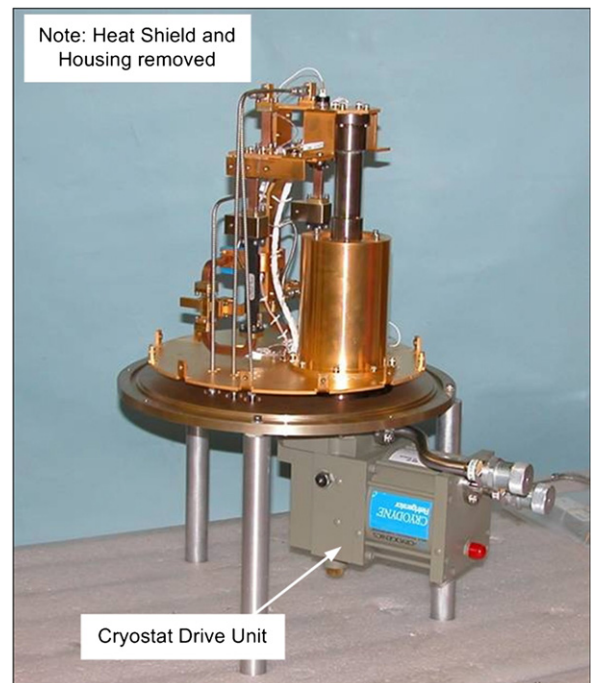


Fig. 7. Cryogenic LNA assembly.

chain. Flexibility for fault isolation is accomplished by having 3 different test injection loops: short loop for injection back into the digital receiver assembly, medium loop for injection at the LNA/post amplifier output and long loop for injection at the LNA input.

The physical arrangement of the RF subsystem “front end” boxes and the feed/LNA assembly at the shroud interface is shown in Fig. 9.

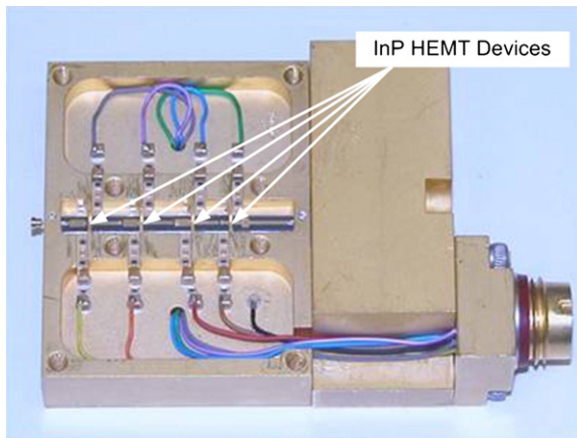


Fig. 8. Ka₂ LNA module.

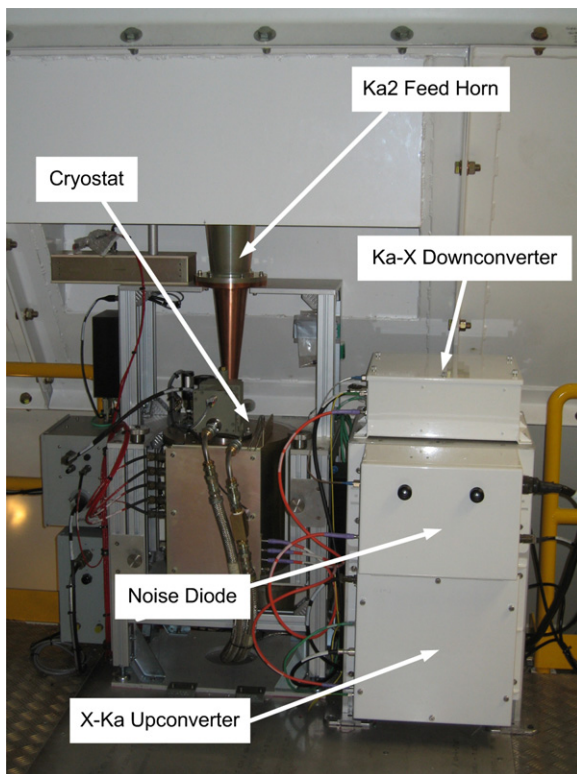


Fig. 9. Ka₂ feed/LNA package.

3.5. Digital receiver subsystem

Demodulation and telemetry processing is done using a “2-box” commercial telemetry subsystem from RT Logic, consisting of a receiver (demodulator/decoder) assembly and a telemetry processor assembly. The demodulation, convolutional (Viterbi) decoding and differential decoding is done in the receiver, while the frame synchronization, Reed Solomon decoding and pseudo noise (PN) decoding is done in the telemetry processor. The front-end of the



Fig. 10. Receiver and test generator racks.

receiver uses high-speed analog-to-digital (A–D) converter technology, with an input bandwidth and sampling rate over 1 GHz. The demodulation and decoding functions are then done using high-speed field programmable gate arrays (FPGAs) or in software, allowing for easy upgrades to different demodulation and decoding types in the future. Custom software development was limited to modifications to the existing DSN monitor and control programs, to reduce development time and cost. The operator interfaces remain basically the same as the existing DSN receiver subsystem, to allow a quick transition from development to transferred operations. This has reduced the required software “soak” period and the required operations personnel training.

The arrangement of the redundant receiver and telemetry processor assemblies in the SPC rack is shown in Fig. 10. The synthesized downconverter that provides the 720 MHz IF signal to both receivers can be seen in the bottom of the Receiver rack.

4. System test results

4.1. Antenna calibration

Since the 34-meter BWG antennas had not been previously tested at the 26 GHz band, it was necessary to do

extensive calibration measurements for each of the 3 antennas. Most of the Ka2 antenna calibration data were taken using a DSN operational system called the Antenna

Table 1
Antenna calibration data—DSS-24.

Parameter	Requirement	Measured	Notes
Pointing error	< 5.5 mdeg	4.0 mdeg	MRE
G/T, 26.0-GHz	> 61.5 dB/K	62.2 dB/K	45-deg elevation
G/T, 25.5–27.0 GHz	> 60.7 dB/K	61.4 dB/K	45-deg elevation
Antenna efficiency	N/A	61.20%	55-deg elevation

Calibration Measurement Equipment (ACME). This system is used for all of the DSN operational bands (S, X, Ka1 and Ka2) and is described in detail in the Deep Space Communication and Navigation Systems Center of Excellence (DESCANSO) series [4]. The details of the antenna calibration results for the Ka2 band are given in [5].

A summary of the principle antenna parameters measured for the Ka2 band is given in Table 1. The pointing error is calculated as the mean radial error (MRE). An example of one of the tipping curves showing the strong dependence of sky temperature (*Tsky*) and system operating temperature (*Top*) with elevation angle for the Ka2 band is shown in Fig. 11. The *T* (antenna microwave), or *Tamw*, is the *Top* with

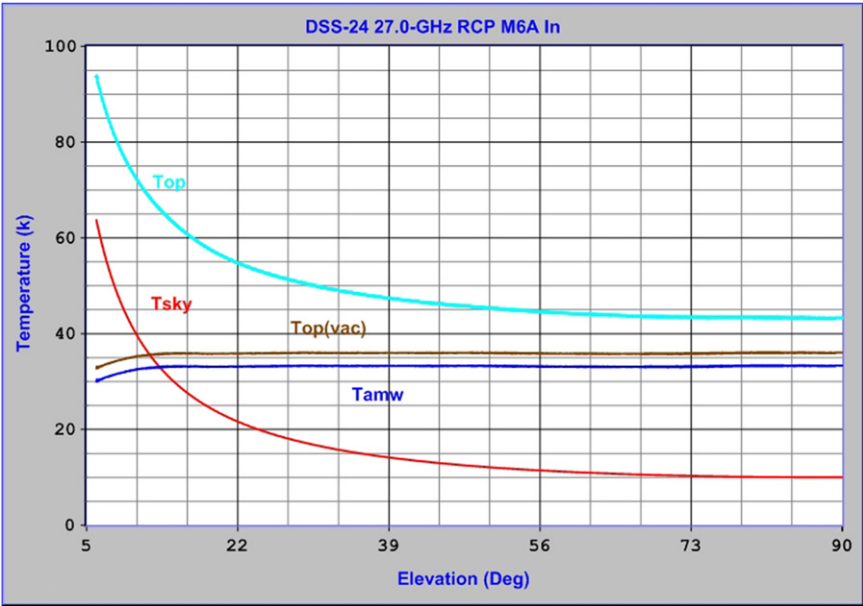


Fig. 11. Ka-band tipping curve at 27 GHz.

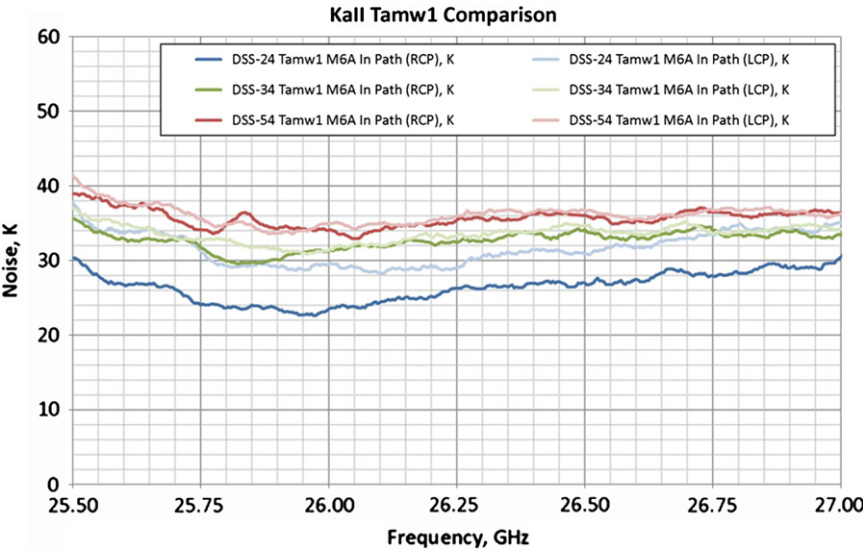


Fig. 12. System noise temperature vs. frequency (vacuum).

atmospheric effects and cosmic background noise taken out. The Top for dry atmospheric conditions, at zenith, is about 45 K. A continuous sweep of system temperature (Tamw, in vacuum) over the Ka2 band was also measured and is shown in Fig. 12 for all 3 antennas. The noise contribution of the M6A dichroic plate, for both RCP and LCP, is shown in Fig. 13.

4.2. Passband flatness

The overall system passband flatness over the nominal 400 MHz telemetry bandwidth was measured using the aperture load as a “flat” noise source. The resulting noise output at the 720 MHz IF, with the input channel set to the Ka2 band center, is given in Fig. 14.

4.3. Phase stability

Although the Ka2 system is not intended to be used for radio science applications, there is a short-term phase stability requirement imposed in order to meet the future Doppler measurement accuracy requirement. This requirement translates to an Allen Deviation requirement of $1.7\text{E-}13$ for a 60 s integration time. The system performance is about an order of magnitude lower than the requirement, as seen in the plot in Fig. 15.

4.4. System linearity

The system linearity was measured, using the ACME system, to be about 0.2% at the -90 dBm noise input level.

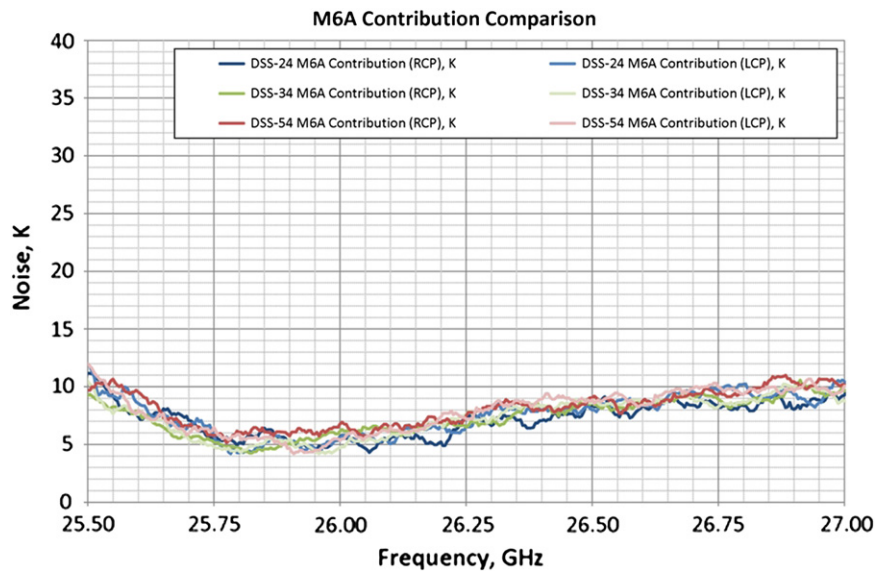


Fig. 13. M6A noise contribution vs. frequency.

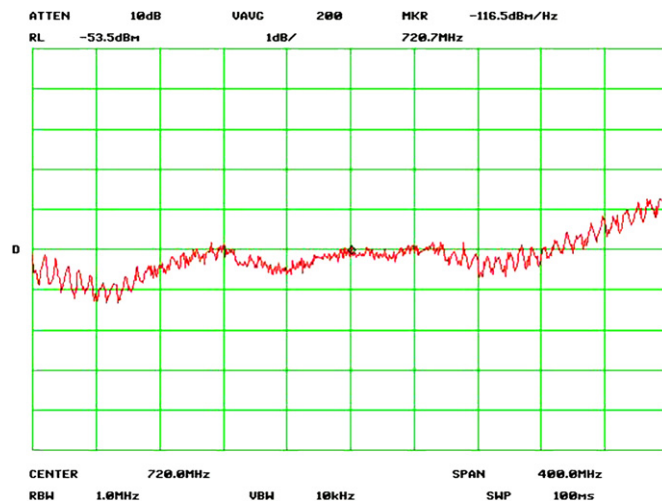


Fig. 14. System passband flatness—400 MHz span (DSS-54).

The compression characteristic was also measured using a spectrum analyzer and is plotted in Fig. 16. The 1 dB input compression point for the high gain mode, at the LNA input, is approximately -63 dBm.

4.5. Bit error rate (BER)

The most fundamental performance of the Ka2 system was characterized by measuring the bit error rate performance. The theoretical reference for rate $\frac{1}{2}$ Viterbi coding is documented in [6]. A summary of measured versus theoretical performance for 100 Mbps data rates is given in Fig. 17, showing good agreement. Based on the measured BER values, the demodulator/decoder loss of the Ka2 system is less than 1 dB.

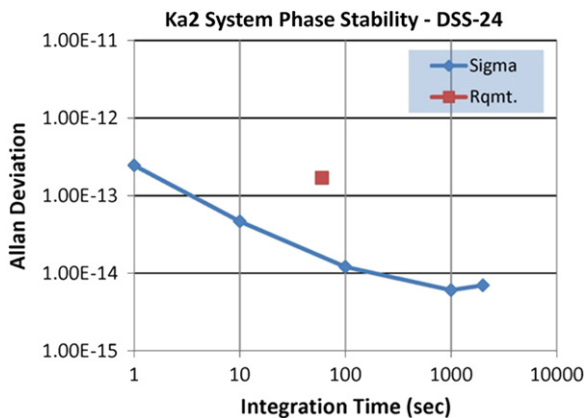


Fig. 15. System phase stability (Allan Deviation, DSS-24).

5. Operations

For ease of operations, the user interfaces to the Ka2 system were designed to resemble the existing S-, X- and Ka1-band interfaces as much as possible. One of the differences is that the SNT measurement is not valid in the presence of a typical strong signal. Rather, the operator is required to measure the SNT before and after a track, when no signal is present, in order to verify that the symbol SNR is within expected limits. Another basic difference is that the Ka2 system uses blind pointing, whereas the S- and X-band systems use CONSCAN tracking and the Ka1 system uses monopulse tracking. The antenna pointing model accuracy is therefore more critical and may need to be updated more often than in the past.

In order to demonstrate the end-to-end performance of the Ka2 system, some “shadow” tracks were run using the LRO spacecraft. These tracks were done on a non-interference basis (NIB) to the primary WS1 ground station. The symbol rate for the LRO Ka-band downlink is 228 Msps with QPSK modulation. For the case of DSS-24 at Goldstone, the ground antenna was near the bore-sight of the spacecraft antenna and the received symbol SNR was very strong, as shown in Fig. 18. The variation over the 50 min pass is less than 1.5 dB, probably due to the variation in system temperature as the spacecraft passes from limb to limb of the moon. The received total power variation (referenced to the COTS receiver input) over the pass is < 1 dB (Fig. 19), showing the good performance of the DSN antenna blind pointing model.

6. Conclusion

The Ka-band Phase 2 upgrade is a major addition to the capability of the DSN, allowing the network to support a

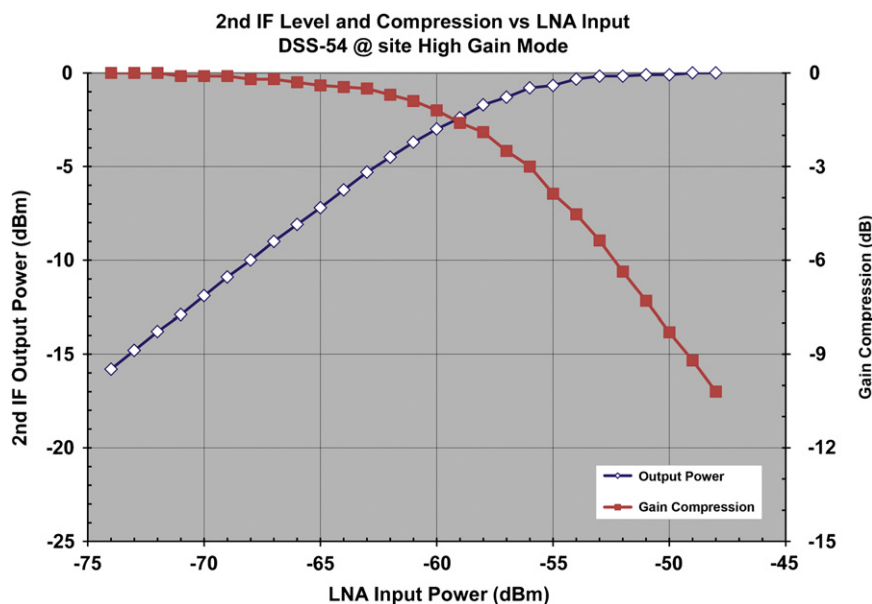


Fig. 16. System linearity—high-gain mode.

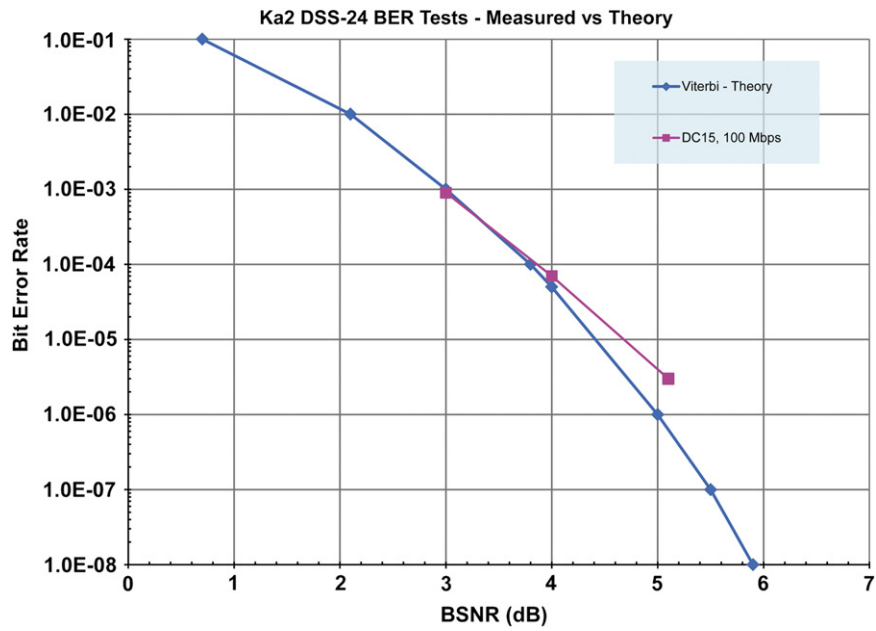


Fig. 17. Bit error rate (100 Mbps)—Measured vs. theory (DSS-24).

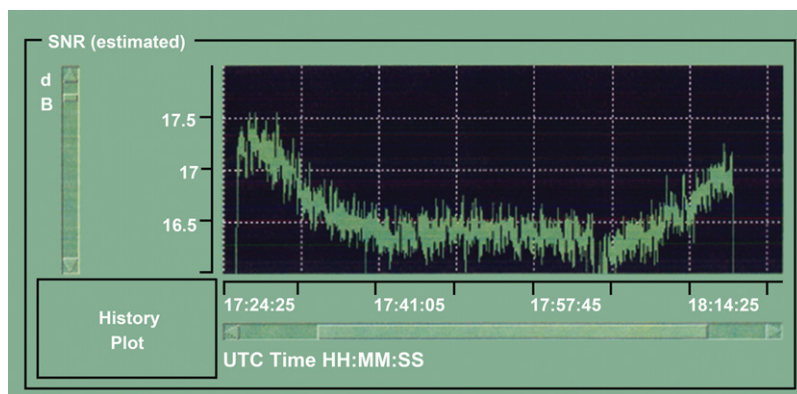


Fig. 18. LRO track—receive symbol SNR (dB) (DSS-24).

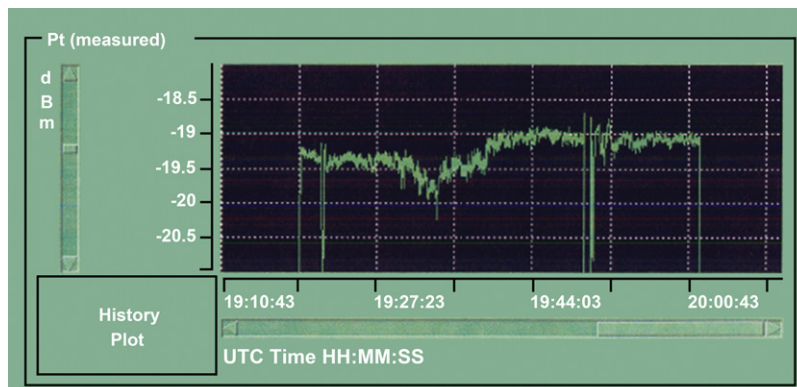


Fig. 19. LRO track—receive total power (dBm) (DSS-24).

new mission set with much higher data rates than the traditional deep-space missions. The RF front-end architecture is designed to allow economical addition of multiple back-end receivers to support multiple spacecraft within the same antenna beam. The Ka-band Phase 2 upgrade has been recently implemented into 3 different 34-meter BWG antennas in the DSN, one at each of the complexes in Canberra (Australia), Goldstone (California) and Madrid (Spain), and is ready to support Category A Ka-band missions.

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References

- [1] K. McCarthy, F. Stocklin, B. Geldzahler, D. Friedman, P. Celeste, NASAs Evolution to Ka-Band Space Communications for Near-Earth Spacecraft, in: Proceedings of the SpaceOps 2010 Conference, April 25–30, 2010.
- [2] W.A. Imbriale, Large Antennas of the Deep Space Network, John Wiley and Sons, New Jersey, 2003 (Chapter 8).
- [3] C. Ho, A. Kantak, S. Slobin, D. Morabito, Link Analysis of a Telecommunication System on Earth, in: Geostationary Orbit, and at the Moon: Atmospheric Attenuation and Noise Temperature Effects, Interplanetary Network Progress Report 42-168, October–December 2006, JetPropulsion Laboratory, Pasadena, California, February 15, 2007, pp. 1–22 <http://ipnpr/progress_report/42-168/168E.pdf>.
- [4] M.S. Reid (Ed.), Low-noise Systems in the Deep Space Network, John Wiley and Sons, New Jersey, 2008 (Chapter 7).
- [5] D. Rochblatt, G. Baines, M. Vazquez, I. Sotuela, C. Snedeker, R. LaBelle, J. Schredder, T. Ridgway, Calibration and Performance Measurements of the NASA Deep Space Network Antennas Upgrade for Ka-Band (26 GHz), in: Proceedings of the 4th European Conference on Antennas and Propagation, (EuCAP), April 12–16, 2010.
- [6] J.H. Yuen (Ed.), Deep Space Telecommunications Systems Engineering, Plenum Press, New York, 1983, p. 225.



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David Rochblatt is the Antenna and Microwave System Engineer for the JPL-DSN. Mr. Rochblatt is the co-author of the book "Low-Noise Systems in the Deep Space Network", which was published by JPL DSN-CANSO and John Wiley in 2008. In this book, Mr. Rochblatt covers the subjects of "Antenna Calibration" and "Microwave Antenna Holography" as applied to the NASA-JPL-DSN, and presents a good portion of his contributions to the science and telemetry advancement of the DSN enabled by these technological developments. In 1995 he received the NASA Exceptional Achievement Medal for developing the Microwave Antenna Holography System. He has published 70 papers and received more than a dozen NASA Achievement Awards.